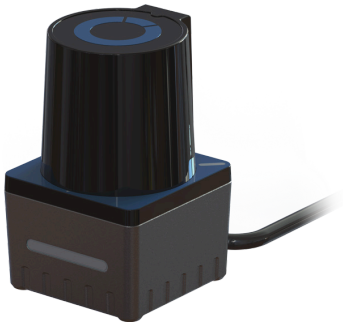


2D SCANNING LIDAR · VB SERIES

180-GRAM OLEI VB SERIES 270° LIDAR

Three variants of a 50 × 50 × 80 mm, 180-gram 270° scanner that draws under 3 watts — the size, weight and power budget of a service robot, with the point rate and the zone logic of a full industrial sensor.

VBD1-10 · VB1-25 · VBD1-25 — the D variants carry the zone-switching I/O; VB1-25 is the point-cloud-only model.



■ WHY THIS SENSOR

- **180 g in a 50 × 50 × 80 mm body** — a third the mass of the LR-1BS series, for chassis with no room to spare
- **≤ 3 W at 12–26 V DC** — battery-platform power budget, no fan, IP65
- **Up to 45 000 points per second** — 27K @ 15 Hz, 36K @ 20 Hz, 45K @ 25 Hz at 0.2° angle resolution
- **0.2–25 m on the -25 variants** — 0.2–10 m on the VBD1-10; reflector range matches the 80% figure on all three
- **< ±20 mm absolute accuracy on the -25 variants** — < ±50 mm on the VBD1-10, ≤ 20 mm repeat throughout
- **16 banks × 3 levels with 4 in / 4 out** — on VBD1-10 and VBD1-25 — 50 mA per output, 200 ms response at 15 Hz
- **M12 12-pin aviation connector plus RJ45** — on the D variants; VB1-25 takes 5.5 mm DC power and RJ45
- **Under-seat or side cable exit, bottom mount** — OleView and SafetyConfigTool on the D variants, ROS driver on VB1-25

0.2–25 m

WORKING RANGE

270°

FIELD OF VIEW

45K

POINTS / SECOND

180 g

WEIGHT

■ APPLICATION SCENARIOS



COMPACT PLATFORMS

Service robots and small AMRs with tight sensor pockets



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



AREA MONITORING

Perimeter and equipment-envelope intrusion detection

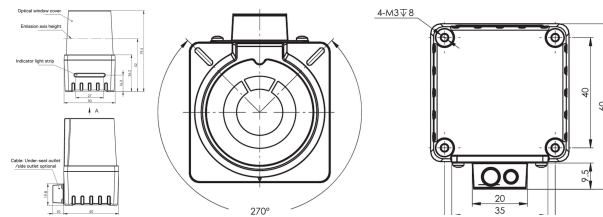


OEM INTEGRATION

Volume platforms where size, weight and power are budgeted

PARAMETER	VBD1-10	VB1-25	VBD1-25
Sensing technology	2D scanning ToF LiDAR		
Laser wavelength	905 ± 20 nm		
Laser safety level	Class 1 (IEC 60825-1:2014)		
Laser channels	1		
Field of view (horizontal)	270°		
Beam divergence	6 mrad		
Scanning frequency	15 / 20 / 25 Hz		
Angle resolution	0.2°		
Working distance (based on reflectivity)	Reflector / 80% 0.2–10 m 10% 0.2–8 m · 4% 0.2–4 m	Reflector / 80% 0.2–25 m 10% 0.2–8 m · 4% 0.2–5 m	Reflector / 80% 0.2–25 m 10% 0.2–10 m · 4% 0.2–5 m
Accuracy ①	Absolute < ±50 mm Repeat ≤ 20 mm	Common < ±20 mm · Reflector < ±40 mm Repeat ≤ 20 mm	Common < ±20 mm · Reflector < ±40 mm Repeat ≤ 20 mm
Distance resolution	1 mm		
Scanning rate	27K @ 15 Hz · 36K @ 20 Hz · 45K @ 25 Hz		
Ethernet connection	100 Mbps Ethernet		
Interface type	Type A M12 12-pin aviation female + RJ45 Ethernet port	5.5 mm DC power interface + RJ45 Ethernet port	Type A M12 12-pin aviation female + RJ45 Ethernet port
Horizontal error (horizontal angle)	≤ ±1°		
Response time	200 ms @ 15 Hz	—	200 ms @ 15 Hz
Setting area	16 banks, 3 levels/bank	—	16 banks, 3 levels/bank
Switching inputs	4	—	4
Switching outputs	4	—	4
Output current	50 mA	—	50 mA
Mechanical dimensions	50 × 50 × 80 mm		
Power consumption @ 25 °C	≤ 3 W (without load)	≤ 3 W	≤ 3 W (without load)
Operating voltage	12–26 V DC		
Weight	180 g		
Degree of protection	IP65		
Operating temperature	–10 to +50 °C		
Storage temperature	–30 to +70 °C		
Relative humidity	< 95% RH		
Indicator status	RGB × 4		
Ambient light limit	80 000 lux		
Installation	Bottom		
Basic software	OleView / SafetyConfigTool	OleView / ROS driver	OleView / SafetyConfigTool

■ OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008. Laser safety Class 1 per IEC 60825-1:2014.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. A dash means the parameter does not apply to that variant. VB1-25 has no switching I/O and no zone configuration — specify VBD1-10 or VBD1-25 where the platform needs bank switching.

Installation. Bottom mount only. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Choose the under-seat or side cable exit at order time. Keep the optical window cover clear of spray and dust and clean it on a schedule in dirty environments.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-VBSERIES-2026.09 · Source: OLEI VB Series 2D LiDAR Sensor datasheet, DOC.# DSEN-VB Series-202505.